

OLIVER HIXON-FISHER

PhD Researcher | Human Motion Synthesis, Computer Vision & Embodied AI

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RESEARCH PROFILE

Final-year PhD researcher specialising in real-time generative and reconstructive models of human motion for embodied interactive systems. Research spans latent diffusion for reactive human-avatar interaction, pose-centric motion synthesis, sparse-sensor full-body reconstruction, self-supervised representation learning, efficient sequence models and online domain adaptation. Experienced with motion capture, VR/XR sensing, 3D avatars and research systems designed for real-time deployment. Thesis submission expected 19 November 2026, viva anticipated Q1 2027; seeking postdoctoral roles for a Q2 2027 start.

RESEARCH EXPERIENCE

PhD Researcher - Kingston University London | London, UK

2022 - Present

- Develop generative and reconstructive human-motion models spanning diffusion models, GANs, VAEs, Mixture-of-Experts, Transformers, JEPA, Mamba/state-space models and test-time adaptation.
- Proposed a latent diffusion framework for continuous reactive motion synthesis from human input, including interaction-contact supervision and a real-time webcam-to-motion pipeline for VR/AR/robotics applications.
- Developed Sparse2Motion, a compact full-body pose reconstruction system from HMD and controller signals using a JEPA encoder, looped Transformer with adaptive computation, and Mamba-based online adaptation; manuscript currently under review at Intelligent Systems with Applications (Elsevier).
- Built the Kingston VR-MoCap dataset: paired Meta Quest sensor trajectories, Rokoko Smartsuit Pro II full-body motion capture and RGB validation footage for studying the pseudo-VR to real-VR domain gap.
- Managed a two-person R&D team on a seedcorn project for an e-commerce prototype using webcam-driven real-time 3D avatar manipulation and body-dimension estimation.

EDUCATION

PhD, Machine Learning / Human Motion - Kingston University London | London, UK

Expected 2027

Thesis: A novel method of generating real-time dynamic pose-based natural motion. **Supervisors:** Dr Jarek Francik and Prof Dimitrios Makris. Thesis submission: 19 November 2026; viva anticipated Q1 2027.

TECHNICAL EXPERTISE

Generative & representation learning: diffusion models; GANs; VAEs; Mixture-of-Experts; JEPA; self-supervised learning. **Sequence modelling & adaptation:** Transformers; looped Transformers; adaptive computation; Mamba/state-space models; test-time training; domain adaptation. **Human motion & embodied systems:** motion synthesis; 3D pose reconstruction; reactive interaction; SMPL; AMASS; motion capture; VR/XR sensing; avatars; real-time inference. **Evaluation:** FID and diversity metrics; MPJPE; MPJVE; MPJRE; jitter; model-size and inference-time profiling.

PUBLICATIONS

Peer-reviewed and accepted work

Hixon-Fisher, O., Francik, J., & Makris, D. *High-Fiving with the Machine: Synthesising Reactive Motion from Human Input with Interaction Contact Labels*. 18th Annual ACM SIGGRAPH Conference on Motion, Interaction and Games (MIG 2025), ACM, long paper, Article 13, pp. 13:1-13:11, 2025, DOI - 10.1145/3769047.3769063

Hixon-Fisher, O., Francik, J., & Makris, D. *Motion Synthesis via Adaptive Guidance Experts*. In Computer Vision, Imaging and Computer Graphics Theory and Applications, Communications in Computer and Information Science, Springer Nature, 2025. Accepted / in press.

Hixon-Fisher, O., Francik, J., & Makris, D. *Pose-Centric Motion Synthesis Through Adaptive Instance Normalization*. Proceedings of the 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2025), Vol. 2: VISAPP, SCITEPRESS, pp. 39-47, 2025, DOI - 10.5220/0013102100003912

Manuscripts under review and preprints

Hixon-Fisher, O., Francik, J., & Makris, D. *Shaking Hands with the Machine - Reactive Motion Synthesis*. SSRN preprint, posted 4 June 2026. Extended journal version currently under review at Sensors (MDPI).

Hixon-Fisher, O. et al. *Sparse2Motion: Adaptive Full-Body Pose Reconstruction from VR Inputs*. Under review at Intelligent Systems with Applications (Elsevier), 2026.

Earlier research output

Hixon-Fisher, O. *A Combined GAN Approach Towards Procedural Generation in 3D Video Game Level Design*. 2nd International Conference on Interactive Media, Smart Systems and Emerging Technologies - conference poster based on master's dissertation, 2022.

TEACHING, LEADERSHIP & ACADEMIC ACTIVITY

- Teaching Assistant: Machine Learning and Artificial Intelligence; Machine Learning and Deep Learning, Kingston University.
- Speaker, Kingston ECE Postgraduate Research Conference (2022, 2024, 2025).
- 3rd Prize, Oral Presentations (Judges' Choice Awards), Kingston ECE PGR Conference, 2025.
- 2025 talk: "Shaking Hands With The Machine: A Novel Method of Synthesising Reactive Motion From Human Input."

SELECTED RESEARCH CONTRIBUTIONS

- Reactive motion synthesis: real-time latent diffusion for open-ended human-avatar interaction, with contact-aware supervision and webcam-to-motion domain adaptation.
- Sparse-to-full-body reconstruction: compact VR pose reconstruction from three tracked devices using self-supervised JEPA representations, adaptive computation and online Mamba adaptation.
- Efficient motion generation: adaptive instance normalisation and Mixture-of-Experts methods for temporally coherent pose-centric synthesis on CMU MoCap and HumanAct12.
- Data collection: paired real VR sensor and full-body mocap capture to evaluate the deployment-domain gap that is hidden by pseudo-VR benchmarks.

SELECTED PRESENTATION & RECOGNITION

Shaking Hands With The Machine: A Novel Method of Synthesising Reactive Motion From Human Input - Kingston ECE PGR Conference	2025
• Awarded 3rd Prize for Oral Presentations (Judges' Choice Awards).	

REFEREES

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